INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science and Engineering

End-Term Examination

Course: B.Tech

Semester: 1st

Branch: CSE

Subject Code: CSE-1004
Subject Name: FoEE

Max. Marks: 60
Time: 03hr

NOTE:

1. **All the questions are compulsory.**
2. **While solving write question number and its part clearly.**
3. **Solve questions in given sequence and all the sub parts should be at same place.**

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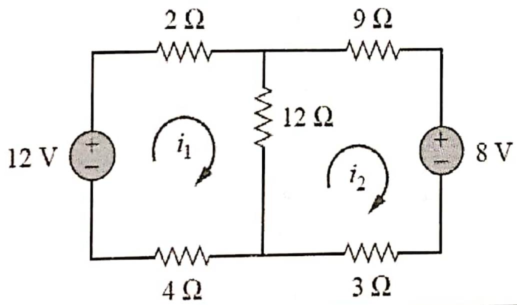
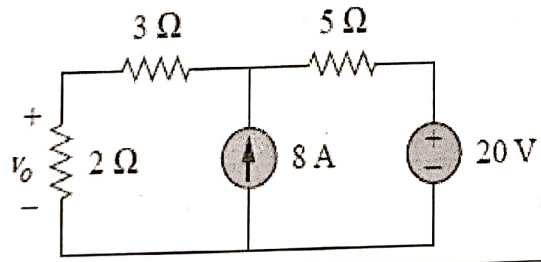
1. A An electrical system comprises voltage sources and loads, as illustrated in Figure. Analyze the circuit, identify the possible meshes, and determine the current flowing through each mess currents i_1 and i_2. **6**

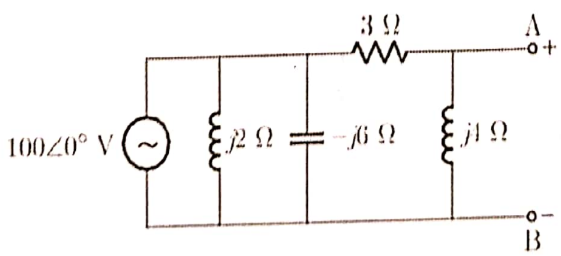


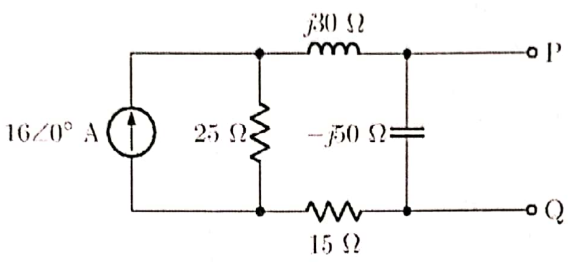
1. B Consider an electrical distribution system in a prototype project, consisting of one voltage source, one current sources, and three loads between the nodes, as illustrated in Figure. Apply the superposition theorem and find the voltage across the 2 Ω load. **6**



2. A An AC electrical network consists of a voltage source connected to a combination of resistor, inductor, and capacitor components, as shown in the figure. Apply the maximum power transfer theorem, determine: **6**

(a) The load impedance Z_L that maximizes power transfer.
(b) The maximum power delivered to the load.



2. B	An AC electrical network consists of a voltage source connected to a combination of resistor, inductor, and capacitor components, as shown in the figure. Using Norton's theorem, determine: (a) The Norton equivalent current (I_N) (b) Norton equivalent impedance (Z_N) 	6
3. A	A 1000 kVA transformer has primary and secondary turns of 400 and 100 respectively and induced voltage in the secondary is 1000 V. Find (i) the primary voltage (ii) the primary and secondary full load current and (iii) the secondary current when 100 kW load at 0.8 p.f. is connected at the output.	6
3. B	The open circuit (O.C.) and short circuit (S.C.) test data are given below for a single phase, 5 kVA, 200V/400V, 50Hz transformer. O.C test from Low Voltage side: 200V 1.25A 150 W S.C test from High Voltage side: 20VV 12.5A 175W Draw the equivalent circuit of the transformer (i) referred to LV side and (ii) referred to HV side inserting all the parameter values.	6
4. A	Let's consider a diode, resistor, and a step-down transformer (10:1) to construct a half-wave rectifier. Assume a sinusoidal signal, following the Indian standard used for home appliances, as the input. Calculate the following: a) Draw the electrical circuit of the half-wave rectifier and output waveform b) Calculate the average DC voltage, RMS value, efficiency, peak inverse voltage, and frequency of the output waveform.	6
4. B	Explain the construction and working principle of an NPN transistor connected in a common-base configuration, and describe its input and output characteristics.	6
5. A	Convert the following number system: (a) Convert $(657)_8$ into decimal. (b) Convert $(2348)_{10}$ into hexadecimal (c) Convert $(67A9)_{16}$ into octal	6
5. B	Implement all the basic gates using the NAND and NOR gates.	6